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SKILLS

Languages: Python, JavaScript
Frameworks: Django, Flask, FastAPI, GraphQL, Django Rest API, Streamlit, Tornado, HTML, CSS, JQuery, Ajax
Technology: Git, AWS, PostgreSQL, Docker, Alembic, MySQL, SQLAlchemy, Redis, Celery, MongoDB, Pytest, Selenium, WebSocket, Socket.io, Treafig, Jira, Pandas, Numpy, Nginx, Apache

WORK EXPERIENCE

THOUGHTMINDS | SENIOR SOFTWARE ENGINEER

India | 2025 January - Present

1. Integrated **PostGIS** and **GeoAlchemy** for advanced location-based search functionality.
2. Monitored and optimized API performance using **Azure Log Analytics**.
3. Automated data ingestion and transformation pipelines via **Azure Data Factory**.
4. Contributed to an AI-based trade finance automation platform for document and compliance processing.
5. Implemented hub-and-spoke **multi-agent** orchestration using **Autogen** and **MCP**, enabling role-based collaboration and compliance-focused automation in KYC, fraud detection support, and customer engagement copilots.
6. Created modular agents using **LangGraph**, **LangChain**, and **OpenAI**, supporting use cases such as RAG, reasoning chains, and autonomous tool-based operations in complex, multi-turn workflows.
7. Built operational controls around AI query pipelines, including SQL audit trails, query explainability, and risk-aware execution monitoring.

NOVIGO SOLUTIONS | SENIOR SOFTWARE ENGINEER

India | 2024 September - 2024 December

1. Developed a module to auto-generate **Pydantic** models from JSON data, streamlining data validation and serialization processes, with support for complex nested structures and seamless integration into existing systems.

GENPRO RESEARCH | SOFTWARE ENGINEER L2

India | 2024 June - 2024 August

1. Implemented **WebSockets** for real-time communication in web applications using **Tornado**, and **Socket.IO** for bidirectional communication between clients and servers, enhancing interactive web applications with live updates and event-driven functionality.
2. Designed and developed **API architecture**, improving system flexibility, scalability, and performance by defining endpoints, request/response formats, authentication methods, and data validation.
3. Proficient in **Bash scripting**, with experience in automating tasks, creating custom tools, and improving workflow efficiency.
4. Implemented **Docker** and **Docker Compose** to containerize multiple applications, reducing deployment times by 50% and ensuring consistent environments across all stages of development and production.
5. Extensively utilized **Celery** and **Celery Canvas** for task execution, messaging, and job scheduling, implementing complex workflows for efficient process management.
6. Automated the classification of medical graphs using **OpenCV**, improving data handling and identification accuracy.
7. Implemented distributed locking mechanisms using **Redis** to prevent redundant external API calls in a microservices architecture.
8. Built an agent-based table summarizer with **OpenAI** and **LLM** that generates concise summaries from complex datasets, optimizing data analysis and accuracy.

GENPRO RESEARCH | SOFTWARE ENGINEER L1

India | 2022 June - 2024 May

1. Developed real-time communication and streaming functionality using **SSE**(Server Sent Events) and **FastAPI**, improving response time and user experience.
2. Integrated **Microsoft SSO** for authentication, reducing login friction.
3. Created Python modules using **Beautiful Soup** to parse tables and **Slate.js** to convert docx files to JSON and vice versa.
4. Developed Python module for converting docx files to **slate.js** compatible JSON data and vice versa using **python-docx** package.
5. Deployed **Traefik** as a reverse proxy, optimizing load balancing and traffic routing.
6. Configured **Nginx** for static asset delivery, streamlining front-end resource loading and shortening page load times.
7. Automated database tasks with **PostgreSQL cron** jobs and custom functions, ensuring data consistency.
8. Built a comprehensive testing framework with **Pytest**, which streamlined the testing process and increased code coverage by 20%, ensuring high-quality software releases.

ACTIONFI TECHNOLOGIES | PYTHON DEVELOPER

India | 2020 Dec - 2022 May

1. Secured file transfers and maintained data integrity by implementing **SFTP**(Secure File Transfer Protocol) with the **Paramiko** package.
2. Established a high-performance asynchronous task queue leveraging **Celery** and **RabbitMQ**, allowing for effective handling of up to 15 distinct workflows simultaneously, boosting team productivity and accelerating project turnaround times.
3. Performed data transformations with **Numpy** and **Pandas**, delivering accurate results aligned with client requirements.
4. Crafted an end-to-end testing strategy with **Selenium**, resulting in a tangible 25% decrease in testing time and enabling faster feedback loops for developers, thus accelerating project timelines.
5. Ensured application stability by writing unit and integration tests using **Django's testing framework**.

PROJECTS

E CART

PYTHON, DJANGO, JAVASCRIPT, AJAX, AWS EC2, MYSQL, TWILIO, SSO

- Launched a referral program integrated into the e-commerce platform, boosting customer acquisition rates by 15% and driving a significant increase in revenue through incentivized purchases.
- Integrated **Google Authentication**, enabling seamless sign-in for users while increasing customer satisfaction scores by 20%; the authentication process is now utilized by more than 70% of the active user base.
- Devised and incorporated **OTP** (One-Time Password) authentication using **Twilio**, establishing an additional security layer that reduced account breaches and reinforced trust in the system's integrity.

PROJECT CENTER MANAGEMENT

PYTHON, DJANGO, HTML, CSS, JAVASCRIPT, AJAX, JQUERY, SQLITE

- Engineered a comprehensive full-stack Django application that integrated project management, attendance tracking, and leave management functionalities; achieved a 30% reduction in administrative workload for department heads and managers, improving overall efficiency.

NOTABLES AND ACHIEVEMENTS

- Created a DSUR Automation prototype that optimized documentation processes, saving 30 hours per month for the regulatory affairs team and boosting efficiency. [\[Link\]](#)

EDUCATION

Diploma In Computer Engineering

GOVERNMENT POLYTECHNIC COLLEGE MANANTHAVADY

Wayanad, Kerala | 2018-2020